

3.1 Dataset

We use the Shengtao/recipe dataset from Hugging Face, a CSV of 32,722 recipes collected from Allrecipes. After removing 3 duplicate URLs, 32,719 recipes remain. Each recipe has four text fields that we search: a title, a short description, an ingredient list and the directions. The remaining columns are metadata. We use some of them as search filters, store others without using them, and drop the columns that are almost empty (Table 1). The dataset also links an image for most recipes, but this project uses text only.

Table 1. Columns in the dataset and how we use them.

Use	Columns
Searched	title, description, ingredients, directions
Filters	category, rating, rating count, prep / cook / total time, servings, calories, protein, fat, carbohydrates, sodium
Stored, not filtered	author, URL, image URL, review count, yields, other nutrition values
Dropped	13 nutrition columns that are over 99% empty, instructions_list (repeats the directions)

The recipes are short. A typical recipe has 9 ingredient lines, a 26-word description and 95 words of directions (Table 2). They fall into 22 categories, but not evenly. Appetizers and snacks, desserts, side dishes and world cuisine together make up 43% of the data, while the smallest categories have only one or two recipes.

Table 2. Length of the text fields and total time, over 32,719 recipes.

Field	Median	10th percentile	90th percentile
Ingredient lines	9	5	15
Description words	26	11	56
Direction words	95	41	185
Total time (minutes)	50	15	210

The text fields are complete: no recipe is missing its title, description or directions. The metadata has gaps. Cook time is missing for 20% of recipes, and 2,444 recipes that were never rated store a rating of 0 rather than an empty value.